# Kirchhoff calc report

*IEC-based, illustrative values, not for construction. Every number comes from the deterministic engine; text was checked against the digit allowlist.*
*Demo workspace - all companies, people and data are fictional.*

## Summary

All evaluated checks are within limits following applied remediation, leaving no outstanding compliance issues.
Revision r2 · hash 187be57dab13b7fa339de5be0b9d2c2cdd21f7e6e81b70ebf7f42ef85779d868 · fail 0, n/a 2, pass 140, warn 0

## Requirements (tender citations; verification shown per row)

| Id | Clause | Page | Value | Unit | Verified |
|---|---|---|---|---|---|
| REQ-max_lead_time_weeks | §2.3 | 2 | 26 | weeks | yes |
| REQ-rack_count | §3.1 | 3 | 48 | racks | yes |
| REQ-rack_rows | §3.1 | 3 | 6 | rows | yes |
| REQ-racks_per_row | §3.1 | 3 | 8 | | yes |
| REQ-rack_power_kw | §3.1 | 3 | 132 | kW | yes |
| REQ-heat_rejection_liquid_pct | §3.1 | 3 | 80 | % | yes |
| REQ-heat_rejection_air_pct | §3.1 | 3 | 20 | % | yes |
| REQ-auxiliary_load_mw | §3.2 | 3 | 0.30 | MW | yes |
| REQ-supply_voltage_kv | §3.3 | 3 | 33 | kV | yes |
| REQ-supply_frequency_hz | §3.3 | 3 | | | no |
| REQ-source_fault_level_mva | §3.3 | 3 | 1,000 | MVA | yes |
| REQ-source_rx_ratio | §3.3 | 3 | | | no |
| REQ-distribution_voltage_v | §3.3 | 3 | 415 | V | yes |
| REQ-design_power_factor | §3.3 | 3 | 0.95 | | yes |
| REQ-critical_power_redundancy | §4.2 | 4 | 2N | | yes |
| REQ-transformer_loading_normal_pct | §4.3 | 4 | 80 | % | yes |
| REQ-transformer_loading_failure_pct | §4.3 | 4 | 100 | % | yes |
| REQ-ups_battery_autonomy_min | §4.5 | 4 | 5 | minutes | yes |
| REQ-voltage_drop_max_pct | §4.6 | 4 | 2.0 | % | yes |
| REQ-pipework_tier_elevation_m | §5.1 | 5 | 2.6 | m | yes |
| REQ-busway_tier_elevation_m | §5.1 | 5 | 3.2 | m | yes |
| REQ-mechanical_mains_tier_elevation_m | §5.1 | 5 | 3.8 | m | yes |
| REQ-coolant_supply_temperature_c | §6.1 | 6 | 30 | °C | yes |
| REQ-coolant_temperature_rise_k | §6.1 | 6 | 10 | K | yes |
| REQ-cdu_redundancy | §6.2 | 6 | N+1 | | yes |
| REQ-pipework_column_clearance_mm | §6.3.4 | 6 | 300 | mm | yes |
| REQ-max_coolant_velocity_m_s | §6.4 | 6 | 2.5 | m/s | yes |
| REQ-redundancy_alternative | App. B | 7 | Block or distributed | | yes |

## Topology options

| Option | Installed MVA | Normal % | Failover % | Cost USD | Losses kW | Max lead (wk) | SPOF |
|---|---|---|---|---|---|---|---|
| 2N | 18.9 | 38.07 | 76.15 | 5,795,160 | 283.15 | 14 | 0 |
| DR | 12.6 | 57.11 | 76.15 | 4,517,600 | 287.6 | 14 | 0 |
| BLOCK | 12.6 | 76.15 | 76.15 | 4,672,160 | 347.42 | 16 | 0 |

## Checker findings

| Rule | Element | Status | Value | Limit | Unit | Clauses |
|---|---|---|---|---|---|---|
| R01 | T-S1 | pass | 76.15 | 100 | % | §4.3 |
| R01 | T-S2 | pass | 76.15 | 100 | % | §4.3 |
| R01 | T-S3 | pass | 76.15 | 100 | % | §4.3 |
| R01 | T-S4 | pass | 76.15 | 100 | % | §4.3 |
| R02 | A-01 | pass | 1,546 | 1,600 | A | |
| R02 | A-02 | pass | 1,546 | 1,600 | A | |
| R02 | A-03 | pass | 1,546 | 1,600 | A | |
| R02 | A-04 | pass | 1,546 | 1,600 | A | |
| R02 | A-05 | pass | 1,546 | 1,600 | A | |
| R02 | A-06 | pass | 1,546 | 1,600 | A | |
| R02 | B-01 | pass | 1,546 | 1,600 | A | |
| R02 | B-02 | pass | 1,546 | 1,600 | A | |
| R02 | B-03 | pass | 1,546 | 1,600 | A | |
| R02 | B-04 | pass | 1,546 | 1,600 | A | |
| R02 | B-05 | pass | 1,546 | 1,600 | A | |
| R02 | B-06 | pass | 1,546 | 1,600 | A | |
| R02 | FA-01 | pass | 1,546 | 2,500 | A | |
| R02 | FA-02 | pass | 1,546 | 1,600 | A | |
| R02 | FA-03 | pass | 1,546 | 1,600 | A | |
| R02 | FA-04 | pass | 1,546 | 1,600 | A | |
| R02 | FA-05 | pass | 1,546 | 1,600 | A | |
| R02 | FA-06 | pass | 1,546 | 1,600 | A | |
| R02 | FB-01 | pass | 1,546 | 2,500 | A | |
| R02 | FB-02 | pass | 1,546 | 1,600 | A | |
| R02 | FB-03 | pass | 1,546 | 1,600 | A | |
| R02 | FB-04 | pass | 1,546 | 1,600 | A | |
| R02 | FB-05 | pass | 1,546 | 1,600 | A | |
| R02 | FB-06 | pass | 1,546 | 1,600 | A | |
| R03 | FA-01 | pass | 1,600 | 1,600 | A | |
| R03 | FA-02 | pass | 1,600 | 1,600 | A | |
| R03 | FA-03 | pass | 1,600 | 1,600 | A | |
| R03 | FA-04 | pass | 1,600 | 1,600 | A | |
| R03 | FA-05 | pass | 1,600 | 1,600 | A | |
| R03 | FA-06 | pass | 1,600 | 1,600 | A | |
| R03 | FB-01 | pass | 1,600 | 1,600 | A | |
| R03 | FB-02 | pass | 1,600 | 1,600 | A | |
| R03 | FB-03 | pass | 1,600 | 1,600 | A | |
| R03 | FB-04 | pass | 1,600 | 1,600 | A | |
| R03 | FB-05 | pass | 1,600 | 1,600 | A | |
| R03 | FB-06 | pass | 1,600 | 1,600 | A | |

| Rule | Element | Status | Value | Limit | Unit | Clauses |
|---|---|---|---|---|---|---|
| R04 | CB-S1 | pass | 65.42 | 100 | kA | §4.4 |
| R04 | CB-S2 | pass | 65.42 | 100 | kA | §4.4 |
| R04 | CB-S3 | pass | 65.42 | 100 | kA | §4.4 |
| R04 | CB-S4 | pass | 65.42 | 100 | kA | §4.4 |
| R04 | S1 | pass | 65.42 | 100 | kA | §4.4 |
| R04 | S2 | pass | 65.42 | 100 | kA | §4.4 |
| R04 | S3 | pass | 65.42 | 100 | kA | §4.4 |
| R04 | S4 | pass | 65.42 | 100 | kA | §4.4 |
| R04b | A-01 | pass | 52.37 | 85 | kA | §4.4 |
| R04b | A-02 | pass | 43.77 | 85 | kA | §4.4 |
| R04b | A-03 | pass | 45.54 | 85 | kA | §4.4 |
| R04b | A-04 | pass | 47.43 | 85 | kA | §4.4 |
| R04b | A-05 | pass | 49.43 | 85 | kA | §4.4 |
| R04b | A-06 | pass | 50.05 | 85 | kA | §4.4 |
| R04b | B-01 | pass | 52.48 | 85 | kA | §4.4 |
| R04b | B-02 | pass | 52.62 | 85 | kA | §4.4 |
| R04b | B-03 | pass | 54.66 | 85 | kA | §4.4 |
| R04b | B-04 | pass | 53.97 | 85 | kA | §4.4 |
| R04b | B-05 | pass | 51.96 | 85 | kA | §4.4 |
| R04b | B-06 | pass | 50.05 | 85 | kA | §4.4 |
| R04b | FA-01 | n/a | n/a | n/a |  | §4.4 |
| R04b | FA-02 | pass | 65.42 | 85 | kA | §4.4 |
| R04b | FA-03 | pass | 65.42 | 85 | kA | §4.4 |
| R04b | FA-04 | pass | 65.42 | 85 | kA | §4.4 |
| R04b | FA-05 | pass | 65.42 | 85 | kA | §4.4 |
| R04b | FA-06 | pass | 65.42 | 85 | kA | §4.4 |
| R04b | FB-01 | n/a | n/a | n/a |  | §4.4 |
| R04b | FB-02 | pass | 65.42 | 85 | kA | §4.4 |
| R04b | FB-03 | pass | 65.42 | 85 | kA | §4.4 |
| R04b | FB-04 | pass | 65.42 | 85 | kA | §4.4 |
| R04b | FB-05 | pass | 65.42 | 85 | kA | §4.4 |
| R04b | FB-06 | pass | 65.42 | 85 | kA | §4.4 |
| R05 | FA-01 | pass | 0.84 | 2 | % | §4.6 |
| R05 | FA-02 | pass | 1.71 | 2 | % | §4.6 |
| R05 | FA-03 | pass | 1.54 | 2 | % | §4.6 |
| R05 | FA-04 | pass | 1.36 | 2 | % | §4.6 |
| R05 | FA-05 | pass | 1.19 | 2 | % | §4.6 |
| R05 | FA-06 | pass | 1.14 | 2 | % | §4.6 |
| R05 | FB-01 | pass | 0.83 | 2 | % | §4.6 |
| R05 | FB-02 | pass | 0.94 | 2 | % | §4.6 |
| R05 | FB-03 | pass | 0.78 | 2 | % | §4.6 |
| R05 | FB-04 | pass | 0.83 | 2 | % | §4.6 |
| R05 | FB-05 | pass | 0.99 | 2 | % | §4.6 |
| R05 | FB-06 | pass | 1.14 | 2 | % | §4.6 |
| R06 | HALL | pass | 0 | 0 |  | §4.2 |
| R07 | UPS-S1 | pass | 2,157 | 2,500 | kW | §4.5 |
| R07 | UPS-S2 | pass | 2,157 | 2,500 | kW | §4.5 |
| R07 | UPS-S3 | pass | 2,157 | 2,500 | kW | §4.5 |

| Rule | Element | Status | Value | Limit | Unit | Clauses |
|---|---|---|---|---|---|---|
| R07 | UPS-S4 | pass | 2,157 | 2,500 | kW | §4.5 |
| R08 | Z1 | pass | 1,690 | 2,400 | kW | §6.2 |
| R08 | Z2 | pass | 1,690 | 2,400 | kW | §6.2 |
| R08 | Z3 | pass | 1,690 | 2,400 | kW | §6.2 |
| R09 | M-R1 | pass | 157.26 | 157.26 | L/min | §6.1 |
| R09 | M-R2 | pass | 157.26 | 157.26 | L/min | §6.1 |
| R09 | M-R3 | pass | 157.26 | 157.26 | L/min | §6.1 |
| R09 | M-R4 | pass | 157.26 | 157.26 | L/min | §6.1 |
| R09 | M-R5 | pass | 157.26 | 157.26 | L/min | §6.1 |
| R09 | M-R6 | pass | 157.26 | 157.26 | L/min | §6.1 |
| R10 | BR-R1 | pass | 1.13 | 2.5 | m/s | §6.4 |
| R10 | BR-R2 | pass | 1.13 | 2.5 | m/s | §6.4 |
| R10 | BR-R3 | pass | 1.13 | 2.5 | m/s | §6.4 |
| R10 | BR-R4 | pass | 1.13 | 2.5 | m/s | §6.4 |
| R10 | BR-R5 | pass | 1.13 | 2.5 | m/s | §6.4 |
| R10 | BR-R6 | pass | 1.13 | 2.5 | m/s | §6.4 |
| R10 | H-Z1 | pass | 1.26 | 2.5 | m/s | §6.4 |
| R10 | H-Z2 | pass | 1.26 | 2.5 | m/s | §6.4 |
| R10 | H-Z3 | pass | 1.26 | 2.5 | m/s | §6.4 |
| R11 | BR-R1 | pass | 5.8 | 0.3 | m | §6.3.4 |
| R11 | BR-R2 | pass | 3.33 | 0.3 | m | §6.3.4 |
| R11 | BR-R3 | pass | 3.33 | 0.3 | m | §6.3.4 |
| R11 | BR-R4 | pass | 1.8 | 0.3 | m | §6.3.4 |
| R11 | BR-R5 | pass | 4.72 | 0.3 | m | §6.3.4 |
| R11 | BR-R6 | pass | 3.33 | 0.3 | m | §6.3.4 |
| R11 | H-Z1 | pass | 3.8 | 0.3 | m | §6.3.4 |
| R11 | H-Z2 | pass | 1.8 | 0.3 | m | §6.3.4 |
| R11 | H-Z3 | pass | 4.8 | 0.3 | m | §6.3.4 |
| R12 | BOQ | pass | 0 | 0 | lines |  |
| R13 | CDU-1 | pass | 2 | 2 | systems | §6.5 |
| R13 | CDU-2 | pass | 2 | 2 | systems | §6.5 |
| R13 | CDU-3 | pass | 2 | 2 | systems | §6.5 |
| R13 | CDU-4 | pass | 2 | 2 | systems | §6.5 |
| R13 | CDU-5 | pass | 2 | 2 | systems | §6.5 |
| R13 | CDU-6 | pass | 2 | 2 | systems | §6.5 |
| R13 | CDU-7 | pass | 2 | 2 | systems | §6.5 |
| R13 | CDU-8 | pass | 2 | 2 | systems | §6.5 |
| R13 | CDU-9 | pass | 2 | 2 | systems | §6.5 |
| R14 | H-Z1 | pass | 69.3 | 150 | kPa | §6.2 |
| R14 | H-Z2 | pass | 67.04 | 150 | kPa | §6.2 |
| R14 | H-Z3 | pass | 64.79 | 150 | kPa | §6.2 |
| R15 | ACB-1600 | pass | 0 | 26 | weeks | §2.3 |

| Rule | Element | Status | Value | Limit | Unit | Clauses |
|---|---|---|---|---|---|---|
| R15 | ACB-4000 | pass | 0 | 26 | weeks | §2.3 |
| R15 | ACB-5000 | pass | 0 | 26 | weeks | §2.3 |
| R15 | BW-1600 | pass | 0 | 26 | weeks | §2.3 |
| R15 | CBL-240-8P | pass | 0 | 26 | weeks | §2.3 |
| R15 | CDU-1200 | pass | 14 | 26 | weeks | §2.3 |
| R15 | CRB-415 | pass | 0 | 26 | weeks | §2.3 |
| R15 | LVB-415 | pass | 0 | 26 | weeks | §2.3 |
| R15 | MVS-33 | pass | 0 | 26 | weeks | §2.3 |
| R15 | PIPE-DN150 | pass | 0 | 26 | weeks | §2.3 |
| R15 | PIPE-DN200 | pass | 0 | 26 | weeks | §2.3 |
| R15 | TX-3150 | pass | 0 | 26 | weeks | §2.3 |
| R15 | UPS-500 | pass | 0 | 26 | weeks | §2.3 |

## Fixes

R05:FA-01: Replace FA-01 with a parallel cable feeder (Cable feeder 8 x 1C 240 mm2 Cu per phase) (power). Candidate FA-01:cable-CBL-240-8P is chosen because it provides the largest voltage-drop margin per dollar to resolve rule R05 on FA-01. Formula: dU% = 100·sqrt3·I·sum L·(R·cosphi + X·sinphi)·k/U; k = 1 (feeder), (n+1)/(2n) (row busway). Rejected: Upsize FA-01 to 2,000 A busway.
R05:FB-01: Replace FB-01 with a parallel cable feeder (Cable feeder 8 x 1C 240 mm2 Cu per phase) (power). Candidate FB-01:cable-CBL-240-8P is chosen because it yields the largest voltage-drop margin per dollar while clearing rule R05 on FB-01. Formula: dU% = 100·sqrt3·I·sum L·(R·cosphi + X·sinphi)·k/U; k = 1 (feeder), (n+1)/(2n) (row busway). Rejected: Upsize FB-01 to 2,000 A busway.
R11:H-Z2: Drop H-Z2 to pipe_low where it crosses FB-03 (cooling). Candidate H-Z2:local-drop is selected because it achieves the fewest changed pipe segments (least disruption to coordinated routes) to clear rule R11 near FB-03. Formula: no pipe segment on a tier above the busway tier over an electrical run; clearance >= c_min. Rejected: Reroute H-Z2 clear of FB-03.

## Assumptions ledger

[did] 2N: TX-2500 replaced by TX-3150 (R15)
[did] DR: TX-2500 replaced by TX-3150 (R15)
[not_checked] Earth-fault loop impedance (IEC 60364-4-41 §411)
[not_checked] Selectivity between protective devices
[not_checked] Arc flash
[not_checked] Harmonics
[not_checked] Air-side cooling for the air-cooled share of rack heat
[not_checked] MV bus faults
[not_checked] Busway derating above 35 °C ambient (ambient assumed <= 35 °C)
[not_checked] Overload protection (R03) of transformer incomer and UPS breakers: sized to equipment rating, checked by R04 only
[not_checked] Short-circuit withstand of cable feeders
[not_checked] Spoken Gemini Live audio is not digit-filtered; on-screen transcripts are.

## Bill of quantities

| Material | Description | Qty | Unit price USD | Lead (wk) |
|---|---|---|---|---|
| ACB-1600 | ACB 1,600 A frame, Icu 100 kA | 12 | 9,800.00 | 0 |
| ACB-4000 | ACB 4,000 A frame, Icu 100 kA | 8 | 21,000.00 | 0 |
| ACB-5000 | ACB 5,000 A frame, Icu 100 kA | 4 | 26,500.00 | 0 |
| BW-1600 | Busway 1,600 A, 415 V | 487 | 420.00 | 0 |
| CBL-240-8P | Cable feeder 8 x 1C 240 mm2 Cu per phase | 164 | 610.00 | 0 |
| CDU-1200 | Coolant distribution unit 1,200 kW | 9 | 145,000.00 | 14 |
| CRB-415 | Critical (UPS output) board 415 V | 4 | 64,000.00 | 0 |

| Material | Description | Qty | Unit price USD | Lead (wk) |
|---|---|---|---|---|
| LVB-415 | LV main board 415 V | 4 | 85,000.00 | 0 |
| MVS-33 | 33 kV switchboard, 2 incomers + bus-coupler | 1 | 240,000.00 | 0 |
| PIPE-DN150 | Steel pipe DN150 | 68 | 140.00 | 0 |
| PIPE-DN200 | Steel pipe DN200 | 56 | 210.00 | 0 |
| TX-3150 | Cast-resin transformer 3.15 MVA 33/0.415 kV Dyn11 | 4 | 118,000.00 | 0 |
| UPS-500 | UPS module 500 kW, 415 V | 20 | 62,000.00 | 0 |